SET - 2

1)

Evaluate 101^3 using an identity

$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

$(100+1)^3 = 1000000 + 30000 + 300 + 1$

$= 1030301$

2)

If $a+b = 5$ and $ab = 6$, find $a^2 + b^2$

$a^2 + b^2 = (a+b)^2 - 2ab$

$= 25 + 12$

$= 13$

3)

If $a+b = 5$ and $ab = 6$, find $a^3 + b^3$.

$a^3 + b^3 = (a+b)^3 - 2ab(a+b)$

$= 125 - 3 \times 6 \times 5 = 125 - 90$

$= 35$

4)

Find $2^3 + 4^3 + 6^3 + \ldots + 16^3$

$= 2^3(1^3 + 2^3 + \ldots + 8^3) = 8 \times \Sigma n^3$

$\Sigma n^3 = (8 \times 9/2)^2 = 36^2 = 1296$

Sum $= 8 \times 1296$

Ans $= 10368$

5)

Prove: $1^2 + 2^2 + \ldots + n^2 = n(n+1)(2n+1)/6$.

Step 1: (n=1): LHS = 1 RHS $= 1 \times 2 \times 3/6 = 1$, True.

Step 2: Assume true for $n = k$:

$1^2 + \ldots + k^2 = k(k+1)(2k+1)/6$

Step 3: (n=k+1): Add $(k+1)^2$

$= (k+1)[k(2k+1) + 6(k+1)]/6$

$= (k+1)(2k^2 + 7k + 6)/6$

$= (k+1)(k+2)(2k+3)/6$

This is the formula with $n = k+1$

By mathematical induction, the result is true for all n.